

Treat the following exercises:

Exercise I: SOCIM is a real estate company whose main activity is the sale of apartments. It wanted to develop a computerized system for the management of sales.

SOCIM has several buildings. Each building has a name and is located at an address. It contains a number of apartments. Each apartment is characterized by a number, an area, the number of its rooms and a selling price.

When a customer is interested to buy an apartment, he must present himself to the company and provide it by the information about him (identity card number, name, first name, address, telephone, occupation). Then, he had to visit several apartments (not sold). After each visit, we note the date, the remarks given by the customer and its decision. A customer cannot purchase an apartment until visiting it. To acquire an apartment, first it is necessary to establish a promise of sale in which it should note all the information about the desired apartment, including the sale price, as well as information about the purchaser and the advance that he wants to pay. This advance must be greater than 20% of the selling price. The signing of the promise, by the purchaser and sales manager, must occur at a given date, in the presence of a lawyer. Any lawyer is characterized by its last name, first name, address, one, two or three phone numbers and his authorization number.

The purchaser may cancel the sale as long as the final sale contract is not signed. A cancelled promise gives rise to withdraw the sale (abandonment of the sale). Each withdrawal is characterized by its number, its date and the causes of canceling. Otherwise, a contract of sale is drafted by lawyer and signed by the purchaser and sales manager. In the contract, it must include the description of the apartment, the selling price, the type of payment, and the date of signature.

The apartment cannot be delivered to the buyer until only he pays the entire price. During the delivery of the keys, a verbatim (oral report) is stipulated by the commercial Director and signed by the purchaser at a given date.

- 1) Build the **class diagram**. (4 pts)
- 2) Build the **States-transitions** of the object apartment. (3 pts)

Exercise II: Draw the **activity diagram** of the process of funding and launching a project. (3 pts)

When a distributor has a development project or extension of its equipment, it must build a complete file of the project.

Once established, the project file is therefore submitted simultaneously to the Bank and to the Director, which responds very quickly.

If the opinion of the Director is unfavorable, the project is neglected and the Bank is changed.

If the Director agrees to co-finance the project, we expected the response of the Bank to decide to continue or to reconsider the file.

When both answers are positive, a final funding is established and the project is launched.

Exercise III: Students of a training institution want to develop a Web kind social network application called netAtlas. The application allows registered members to create a network of friends and share content. To become a member of netAtlas, a user must register during his visit to the Website. When the administrator validates the registration, the user becomes a member; it can then connect to perform the following operations: search a member, invite a member to be his friend, accept another member as a friend, refuse an invitation of friendship, and remove a friend from the friends list. A member may also publish content. A moderator consults the content published by members and may warn with a message to a member who publishes content not in accordance with the rules of netAtlas; after 3 warnings, the administrator removes this member's account. A member is identified by an email address, a name and a surname. A member may have several friends who are also members of netAtlas. A member may make a publication of content at a given date. A publication is concerned a resource to publish; a resource has a name and can be either a message or a picture/video or a link to a web page. A picture/video in addition to the name, has a size in MB. A link to a Web page has an address (URL).

- 1) Establish the **use case diagram**. (3 pts)
- 2) Establish the **sequence diagram** to the "Publish" use case. (2 pts)
- 3) Establish the **class diagram**. (3 pts)

Exercise IV: It is to automate the borrowing management of cassettes in a video club. To borrow a cassette, a subscriber must express borrower demand to an access rights controller's, by entering the following information: identification card number, name, first name. This request can be accepted or rejected. If it is accepted, the controller displays the authorization message. Otherwise, it gives the reasons for the refusal. In case of acceptance, the controller, at the request of the system, gives the reference of a film. The system then checks the existence of this film in the catalog of films, then the availability of a cassette. As appropriate, it will display an agreement for the request or impossibility. If the response is positive and the participant confirms his request, the cassette will be reserved to him for borrowing.

Prepare the **sequence diagram** for the scenario described above. (2 pts)